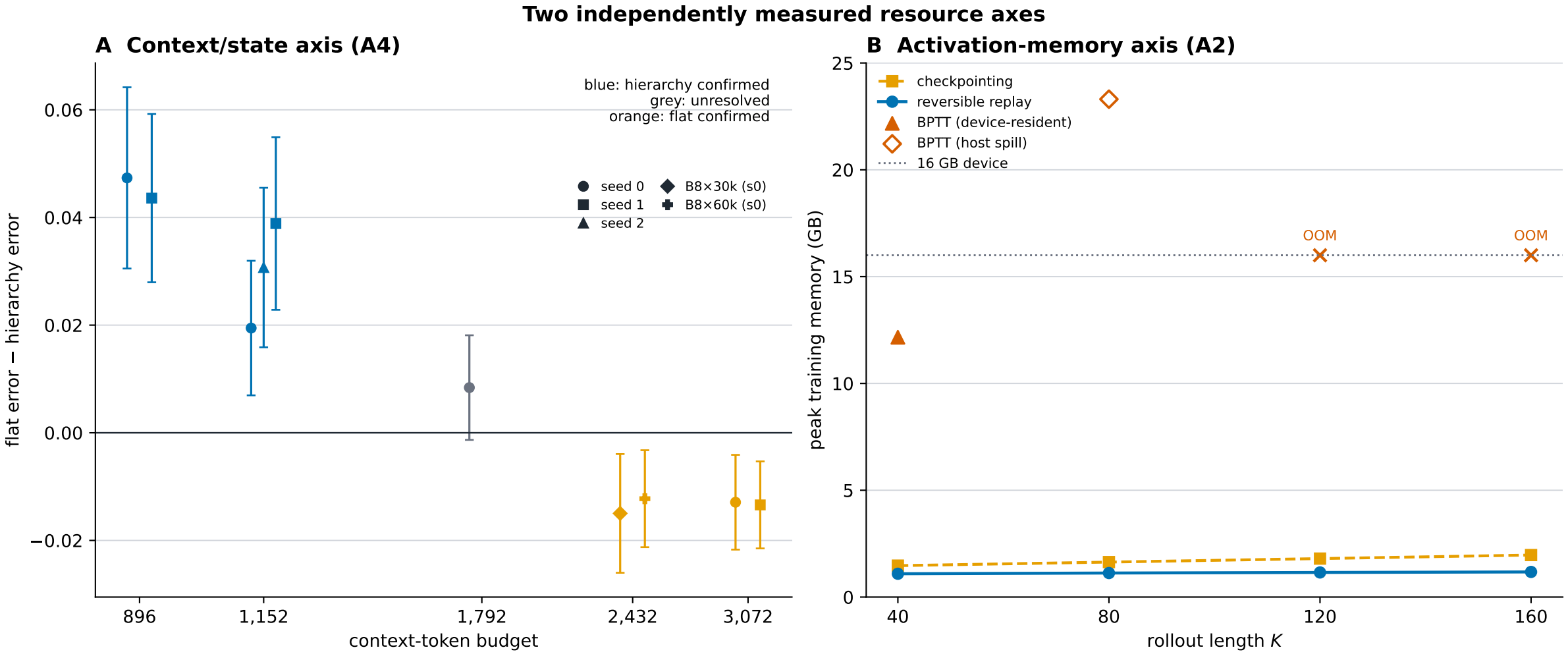
Two independently measured resource axes
A Context/state axis (A4)
blue: hierarchy confirmed
grey: unresolved
orange: flat confirmed
seed 0
seed 1
seed 2
B8×30k (s0)
B8×60k (s0)
flat error − hierarchy error
0.06
0.04
0.02
0.00
−0.02
896
1,152
1,792
2,432
3,072
context-token budget
B Activation-memory axis (A2)
checkpointing
reversible replay
BPTT (device-resident)
BPTT (host spill)
16 GB device
OOM
OOM
peak training memory (GB)
25
20
15
10
5
0
40
80
120
160
rollout length K